WATTWISE — HOUSEHOLD ENERGY HABIT ANALYZER

A PROJECT REPORT

Submitted by

SAJINE SANTHAKUMAR

(2116240701459)

in partial fulfilment of the End Semester Examination for the course

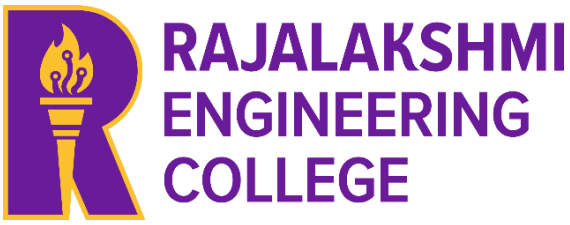
CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF ENGINEERING

COMPUTER SCIENCE AND ENGINEERING



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

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RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled "WATTWISE — HOUSEHOLD ENERGY HABIT ANALYZER" is the bonafide work of SAJINE SANTHAKUMAR (2116240701459), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

SIGNATURE

Dr. J. Manoranjini
HEAD OF THE DEPARTMENT
Department of Computer Science and Engineering
Rajalakshmi Engineering College
Thandalam – 602 105

SIGNATURE

Dr. J. Jinu Sophia
COURSE FACULTY

Department of Computer Science and Engineering
Rajalakshmi Engineering College

Submitted for the End Semester Examination (Individual Project) held on 11 November 2026

INTERNAL EXAMINER

EXTERNAL EXAMINER

DECLARATION

I, SAJINE SANTHAKUMAR (2116240701459), hereby declare that the project report titled "WATTWISE — HOUSEHOLD ENERGY HABIT ANALYZER", submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and integrations — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai
Date: 11 November 2026

SIGNATURE OF THE STUDENT
SAJINE SANTHAKUMAR
2116240701459

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SAJINE SANTHAKUMAR

ABSTRACT

Household electricity consumption is an important aspect of energy management. However, many households monitor electricity usage mainly through monthly electricity bills and manual meter readings. These methods provide limited information about individual appliance consumption, energy-saving activities and maintenance requirements. Maintaining appliance details, tracking meter readings and monitoring service records separately can make household energy management difficult. To address these challenges, the project "WattWise – Household Energy Habit Analyzer" is developed using Zoho Creator, a low-code application development platform.

WattWise provides a centralized application for managing household electrical appliances and monitoring electricity consumption. The application consists of five major forms: Appliance Registry, Meter Reading Log, Energy Audit & Analysis, Energy Saving Action Tracker and Appliance Maintenance & Service. These forms maintain appliance specifications, meter readings, audit information, energy-saving activities and maintenance records. Reports such as list reports, Kanban and calendar views are configured to organize and display the stored information. A dashboard named WattWise Dashboard presents important household energy information through summary panels, charts and embedded reports. Deluge scripting is used as the platform's scripting facility for implementing application logic. Workflows are configured for selected form events and scheduled activities, while a Blueprint is designed to manage the progress of energy-saving actions through stages such as Proposed, Approved, In Progress and Completed. User permissions and role-based access settings are also configured to support controlled access to application data.

The application is developed and evaluated using sample household appliance records, meter readings, energy audit entries and service information. The configured forms, reports, dashboard, workflows and Blueprint demonstrate how a low-code platform can organize household energy information in a single application. WattWise provides a structured approach to recording electricity consumption, monitoring energy-saving activities and maintaining appliance service details.

Keywords: *Low-Code Development; Zoho Creator; Household Energy Management; Deluge Scripting; Workflow Automation; Energy Consumption Monitoring.*

PROJECT AT A GLANCE

Attribute	Details
Project Title	WattWise – Household Energy Habit Analyzer
Application Domain	Household Energy Management
Student Name & Register No.	SAJINE SANTHAKUMAR – 240701459
Zoho Creator Edition	Trial edition (as shown in the application header)
Application Name (Link Name)	WattWise (wattwise)
Published / Live URL	https://creator.zoho.in/appbuilder/sajinesanthakumar.2024.cs628/wattwise/(Development environment)
Demo Credentials (per role)	Admin: sajinesanthakumar.2024.cse@rajalakshmi.edu.in
Number of Forms /	5 / 0
Number of Reports (by type)	5 List Reports, 1 Kanban Report and 1 Calendar Report (Total: 7)
Number of Pages / Dashboards	1 dashboard (WattWise Dashboard)
Number of Workflows	3 – Form-event and scheduled workflows configured
Deluge Functions / Custom Actions	7 Deluge scripts in form workflows / 0 custom functions
Integrations	None external; built-in Send Mail action
Roles & Profiles Defined	Household User and Technician roles; Read and Write permission profiles
Environments Used	Zoho Creator Development Environment
Date of Production Deployment	30 October 2026

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CGPA	Cumulative Grade Point Average
CSE	Computer Science and Engineering
DS	Deluge Script (script identifier)
ER	Entity–Relationship
FR	Functional Requirement
HoD	Head of the Department
NFR	Non-Functional Requirement
REST	Representational State Transfer
TC	Test Case
UAT	User Acceptance Testing
URL	Uniform Resource Locator

CHAPTER 1
INTRODUCTION

1.1 Background

Household energy management is a vital component of sustainable living and cost control. In many homes, energy usage is tracked through overall utility bills and sporadic manual meter readings. Tracking individual electrical appliances, daily electricity consumption, energy-saving practices, and appliance service history across disparate logs and spreadsheets can lead to disorganized record-keeping and difficulties in monitoring household energy habits effectively.

This project addresses that domain through **WattWise — Household Energy Habit Analyzer**, a low-code application developed on **Zoho Creator**. The application is designed to consolidate appliance registries, meter reading logs, energy audits, action trackers, and maintenance schedules into a single platform. It incorporates Deluge workflows, structured reports, and a dedicated dashboard to automate calculations and present comprehensive energy insights.

1.2 Problem Statement

Household energy usage and maintenance data are frequently recorded across fragmented spreadsheets, physical logbooks, and utility receipts. This fragmentation obscures itemized consumption profiles, delays maintenance, and prevents homeowners from assessing energy efficiency initiatives systematic overview.

- Appliance specifications, meter logs, and maintenance histories are stored separately without relational linking.
- Metrics like daily energy consumption, estimated costs, and service schedules are calculated manually, leading to human error.
- Energy-saving initiatives lack lifecycle tracking from proposed stages through execution and completion.

- A unified summary of overall household energy usage, audit results, and active savings actions is not readily accessible.

1.3 Objectives

- To design forms and a linked data model for appliance registry, meter reading logs, energy audits, saving actions, and maintenance records.
- To automate energy consumption calculations, cost estimations, and service reminders using Deluge scripting in form workflows.
- To manage the lifecycle of energy-saving actions through a structured Blueprint and automated workflow notifications.
- To provide organized views across list reports, Kanban boards, calendar views, and a interactive dashboard displaying energy metrics.
- To configure role-based access control and user permission profiles for household residents and maintenance technicians.
- To evaluate all application forms, workflows, reports, and dashboard charts in the development environment.

1.4 Scope of the Project

In scope: five forms (Appliance Registry, Meter Reading Log, Energy Audit & Analysis, Energy Saving Action Tracker, and Appliance Maintenance & Service); form workflows in Deluge; an action item Blueprint; reports; the WattWise Dashboard; role-based access settings; and system evaluation.

Out of scope: direct IoT smart meter integrations, automated billing payments, third-party API connectivity, and custom pro-code portal extensions.

1.5 Justification for a Low-Code Approach

Low-code platforms allow full-featured applications to be developed rapidly using drag-and-drop components and custom logic scripting where needed. This approach fits the requirements of WattWise, as data entry forms, reports, dashboards, and automated reminders can be quickly assembled on Zoho Creator, while specialized logic is easily handled with Deluge scripts.

By abstracting infrastructure and backend hosting overhead, developer effort can be dedicated entirely to refining energy tracking workflows and analytical dashboard features.

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related low-code literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, performance optimisation, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In conventional residential environments, electric power monitoring relies on unautomated workflows or isolated ledger books. Consumption trends are captured primarily through periodic utility billing statements and sporadic direct meter examinations. Storing appliance data, recording meter logs, and tracking maintenance schedules across disconnected documents creates fragmented energy records. Figure 2.1 illustrates this legacy process.

2.2 Limitations of the Existing System

- Appliance specifications, meter logs, and maintenance histories are stored separately without relational linking.
- Metrics like daily energy consumption, estimated costs, and service schedules are calculated manually, leading to human error.
- Energy-saving initiatives lack lifecycle tracking from proposed stages through execution and completion.
- A unified summary of overall household energy usage, audit results, and active savings actions is not readily accessible.

2.3 Low-Code versus Traditional Development

Table 2.1 compares traditional development with the low-code approach for the WattWise application.

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development effort	Custom database schemas, backend calculation scripts, user interfaces, and reporting dashboards must be built manually from scratch.	Forms, reports, and dashboards are assembled using drag-and-drop builders, with Deluge scripting handling automated calculations and workflows.
Skills required	Programming, database and web-development skills.	Platform familiarity and basic Deluge scripting.
Hosting and infrastructure	Servers, deployment and maintenance are managed by the developer.	Hosted by Zoho; the application is accessed through a web link.
Workflow and approvals	State machine logic, scheduled reminders, and notification workflows must be custom-coded.	Action item lifecycles are configured through visual Blueprints and automated form-event workflows.
Security and access	Authentication and authorisation must be designed and coded.	Roles and portal permission profiles are provided by the platform.
Customisation limits	Unlimited flexibility.	Bounded by platform capabilities; Deluge scripts cover energy calculation logic and customized workflows effectively.
Cost	Developer time, server and licence costs.	Developed on a Zoho Creator trial edition; a subscription would be needed for continued use.
Maintenance	Code changes, testing and redeployment.	Changes are made in the builders or in Deluge and tested in the Development environment.

2.4 Literature Survey

The literature survey covers official Zoho Creator documentation, Deluge scripting resources, and technical references for low-code platform capabilities. Table 2.2 provides a summary of the reviewed literature.

Table 2.2: Summary of Literature Survey

S.No.	Author / Source & Year	Title	Key Findings	Relevance to this Project
1	Zoho Corporation, online documentation 30 October 2026	Zoho Creator Help Documentation [2]	Describes the form builder and field types, report types, workflows, approvals, Blueprint, schedules, roles and pages.	Primary reference for building the five forms, reports, workflows and the dashboard.
2	Zoho Corporation, online documentation 30 October 2026	Deluge Script Reference Guide [3]	Describes Deluge syntax, conditions, date and time handling, fetching records, collections and the sendmail task.	Basis for the Deluge scripts DS-01 to DS-07.
3	Zoho Corporation, online learning material 30 October 2026	Zoho Creator Learning Centre [4]	Provides learning material on platform features.	Supporting material for features used in the application.
4	W. Rosam and O. Kalnichevski	Low-Code Development: A Practical Guide [1]	Introduces low-code concepts and platforms.	Supports the low-code justification and the no-code to pro-code discussion.
5	Authors: D. Liu et al.	What's Wrong With Low-Code Development Platforms? An Empirical Study of Low-Code Development Platform Bugs	Identifies common bugs in low-code platforms, including application design issues, unexpected system behavior, and user interface graphics problems	Informs testing workflows, validation rules, energy calculations, and report accuracy for system reliability.

2.5 Emerging Trends in Low-Code Development

Low-code platforms enable rapid development of domain-specific applications like household energy tracking. By integrating visual builders, Deluge scripts, workflow automation, and dashboard analytics, platforms like Zoho Creator provide an optimal environment for building WattWise.

2.6 Proposed System

The proposed system, WattWise – Household Energy Habit Analyzer, consolidates appliance records, meter readings, energy audits, action trackers, and service details across five interconnected forms. Deluge scripts automate consumption calculations, cost estimations, and service reminders, while a Blueprint governs energy-saving action lifecycles. Multi-view reports and a dedicated dashboard provide comprehensive visibility into household energy trends.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Household Administrator	Manage all energy records and review exceptions	Intended administrator
Household User	Record readings, review appliances and actions	Role configuration explored
Technician	Maintain service details	Role configuration explored; login not verified

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Priority
FR-01	Register household appliances	High
FR-02	Store current and previous meter readings	High
FR-03	Calculate consumption and illustrative tariff estimate	High
FR-04	Record appliance-wise energy audits	High
FR-05	Track saving actions and status	High
FR-06	Maintain appliance service records	High
FR-07	Support alerts, reminders and role-based access	Medium
FR-08	Show dashboard summaries and chart	High
FR-09	Support status transitions through Blueprint	Medium

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Usability	Readable forms and navigation
NFR-02	Integrity	Valid dates, readings and numeric values
NFR-03	Security	Permissions matched to role
NFR-04	Maintainability	Named forms and workflows
NFR-05	Performance	Responsive reports at household data scale
NFR-06	Availability	Browser access subject to platform subscription

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator trial/development builder
Client	Internet-connected laptop or desktop
Browser	Modern web browser
Scripting	Deluge editor (execution not fully verified)
Data store	Zoho Creator built-in storage
External hardware	None required

3.5 Use Case Model

A household administrator registers appliances, logs readings, records audits, creates saving actions and reviews reports. A technician may update service records if suitable access is eventually verified.

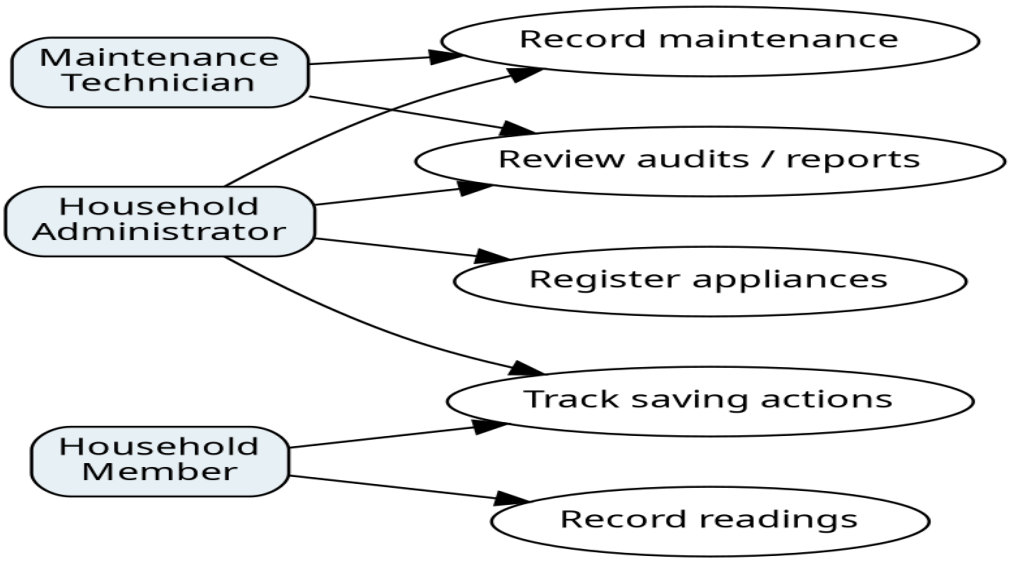


Figure 3.1: Use Case Diagram of the Application

Table 3.5: Use Case Description – Submit and Approve an Internship Application

Attribute	Description
Use case	UC-01: Monitor household consumption
Actors	Household user / administrator
Precondition	Appliance and meter details available
Main flow	Register appliance → enter reading → review consumption → audit → create saving action
Postcondition	Records appear in associated reports
Exception	Invalid or missing reading requires correction

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

The browser interface connects to Zoho Creator forms, reports, dashboard pages and workflow configuration. Zoho Creator stores the records. No independent backend, external database or confirmed API integration is part of the verified design.

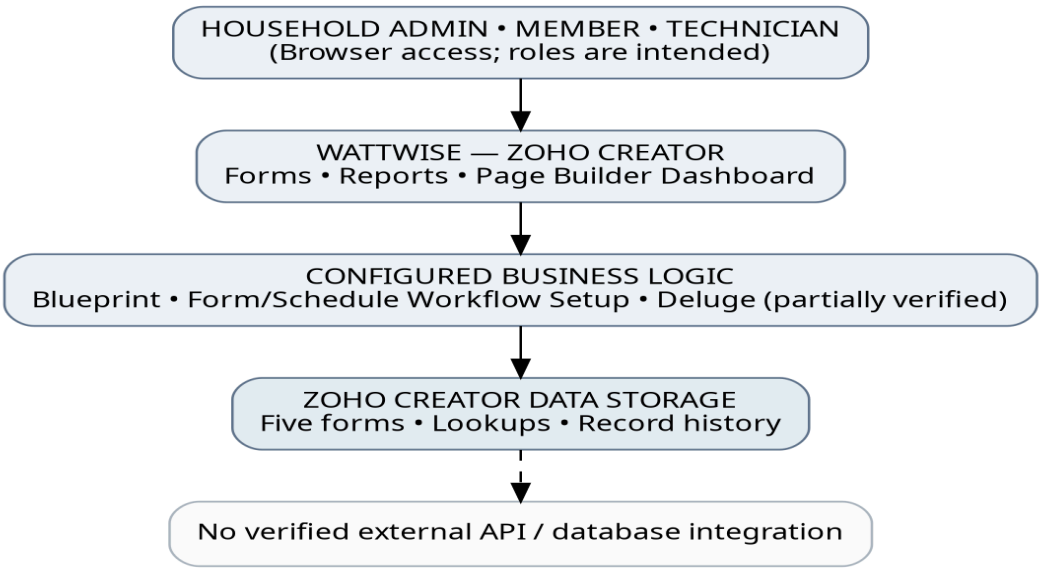


Figure 4.1: System Architecture of the Application

4.2 Data Model

Appliance Registry is the master list for appliance-related records. Energy Audit & Analysis, Energy Saving Action Tracker and Appliance Maintenance & Service refer to appliances using lookup relationships where configured. Meter Reading Log represents household-level measurements rather than individual-appliance meter measurements.

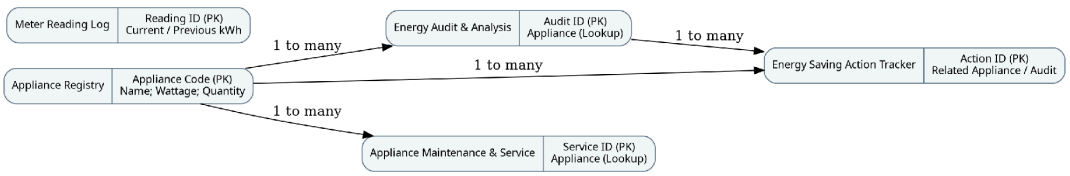


Figure 4.2: Entity–Relationship Diagram

Table 4.1: Entities and Relationships

Entity	Key information	Relationship
Appliance Registry	Code, name, wattage, quantity, usage	Master appliance record
Meter Reading Log	Reading date, previous/current kWh, tariff	Household meter log
Energy Audit & Analysis	Audit dates, appliance lookup, estimated consumption	References appliance
Energy Saving Action Tracker	Action title, priority, status, target date	May reference appliance

4.3 Form Design

The following field specifications consolidate the recorded design. Some field types and link names beyond those explicitly identified in the source notes require final inspection in the builder; the tables therefore describe the business-level fields without asserting unverified mandatory flags.

Table 4.2: Field Specification – Student Registration Form

Field Label	Link Name[1]	Field Type	Mandatory[2]	Validation / Remarks	Field Label
Appliance Name	Appliance_Name	Single Line	To confirm	Name of item	Appliance Name
Appliance Code	Appliance_Code	Auto Number	Auto	System-generated reference	Appliance Code
Appliance Category	Appliance_Category	Drop Down	To confirm	Cooling / Lighting / Kitchen / Entertainment / Laundry / Other	Appliance Category
Household Area	Household_Area	Drop Down	To confirm	Living Room / Bedroom / Kitchen / Bathroom / Study Room / Other	Household Area

Brand	Brand	Single Line	To confirm	Manufacturer	Brand
Model Number	Model_Number	Single Line	To confirm	Model reference	Model Number
Rated Power (Watts)	Rated_Power_Watts	Number	To confirm	Nonnegative rated power	Rated Power (Watts)
Quantity	Quantity	Number	To confirm	Positive integer expected	Quantity
Average Daily Usage (Hours)	Average_Daily_Usage_Hours	Decimal	To confirm	0–24 hours expected	Average Daily Usage (Hours)
Days Used Per Week	Days_Used_Per_Week	Number	To confirm	0–7 days expected	Days Used Per Week
Standby Power (Watts)	Standby_Power_Watts	Decimal	To confirm	Nonnegative standby wattage	Standby Power (Watts)
Energy Efficiency Rating	Energy_Efficiency_Rating	Drop Down	To confirm	Rating entered with appliance	Energy Efficiency Rating
Purchase Date	Purchase_Date	Date	To confirm	Optional purchase record	Purchase Date
Appliance Condition	Appliance_Condition	Drop Down	To confirm	Condition assessment	Appliance Condition
Active Status	Active_Status	Decision Box	To confirm	Active / inactive	Active Status
Notes	Notes	Multi Line	To confirm	Additional description	Notes

Table 4.3: Field Specification – Company Details Form

Field Label	Link Name[1]	Field Type	Mandatory[2]	Validation / Remarks	Field Label
Reading ID	Reading_ID	Auto Number	Auto	Generated identifier	Reading ID
Reading Date	Reading_Date	Date	To confirm	Date of present reading	Reading Date
Reading Type	Reading_Type	Drop Down	To confirm	Daily / Weekly / Monthly	Reading Type
Meter Reading (kWh)	Meter_Reading_kWh *	Decimal	To confirm	Current cumulative reading	Meter Reading (kWh)
Previous Reading (kWh)	Previous_Reading_kWh *	Decimal	To confirm	Previous cumulative reading	Previous Reading (kWh)
Previous Reading Date	Previous_Reading_Date	Date	To confirm	Date used for interval	Previous Reading Date
Meter Photo	Meter_Photo	Image	To confirm	Optional image evidence	Meter Photo
Units Consumed (kWh)	Units_Consumed_kWh	Decimal	To confirm	Current minus previous; automation unverified	Units Consumed (kWh)
Number of Days	Number_of_Days	Number	To confirm	Positive interval	Number of Days
Average Daily Consumption	Average_Daily_Consumption	Decimal	To confirm	Units divided by interval	Average Daily Consumption
Estimated Bill Amount	Estimated_Bill_Amount	Currency	To confirm	Indicative calculation only	Estimated Bill Amount
Tariff per Unit	Tariff_per_Unit	Currency	To confirm	Illustrative tariff assumption	Tariff per Unit
Unusual Consumption Observed	Unusual_Consumption_Observed	Decision Box	To confirm	Flag for manual review	Unusual Consumption Observed
Reading Notes	Reading_Notes	Multi Line	To confirm	Supplementary notes	Reading Notes

Table 4.4: Field Specification – Internship Application Form

Field Label	Link Name[1]	Field Type	Mandatory[2]	Validation / Remarks
Audit ID	Audit_ID	Auto Number	Auto	Generated identifier
Audit Title	Audit_Title	Single Line	To confirm	Audit description
Audit Start Date	Audit_Start_Date	Date	To confirm	Start of comparison
Audit End Date	Audit_End_Date	Date	To confirm	End of comparison
Audit Period	Audit_Period	Drop Down	To confirm	Audit interval label
Audit Status	Audit_Status	Drop Down	To confirm	Progress state
Audit Conducted By	Audit_Conducted_By	Single Line	To confirm	Auditor or administrator
Appliance	Appliance	Lookup	To confirm	Links Appliance Registry
Appliance Wattage	Appliance_Wattage	Number	To confirm	Power used for estimate
Quantity Used	Quantity_Used	Number	To confirm	Units used
Average Usage Hours	Average_Usage_Hours	Decimal	To confirm	Average hours used
Usage Days	Usage_Days	Number	To confirm	Days in audit period
Total Estimated Consumption	Total_Estimated_Consumption	Decimal	To confirm	Appliance estimated energy (kWh)
Meter Consumption	Meter_Consumption	Decimal	To confirm	Observed meter interval (kWh)
Consumption Difference	Consumption_Difference	Decimal	To confirm	Difference for review
Audit Findings	Audit_Findings	Multi Line	To confirm	Findings and exceptions
Recommendations	Recommendations	Multi Line	To confirm	Proposed adjustments

Table 4.5: Field Specification – Internship Attendance Form

Field Label	Link Name[1]	Field Type	Mandatory[2]	Validation / Remarks
Action ID	Action_ID	Auto Number	Auto	Generated identifier
Action Title	Action_Title	Single Line	To confirm	Title of intervention
Related Audit	Related_Audit	Lookup	To confirm	Energy Audit reference
Related Appliance	Related_Appliance	Lookup	To confirm	Appliance Registry reference
Action Category	Action_Category	Drop Down	To confirm	Behaviour change / maintenance etc.
Problem Identified	Problem_Identified	Multi Line	To confirm	Observed issue
Recommended Action	Recommended_Action	Multi Line	To confirm	Action to be taken
Priority	Priority	Drop Down	To confirm	Low / Medium / High
Responsible Person	Responsible_Person	Single Line	To confirm	Assigned action owner
Action Start Date	Action_Start_Date	Date	To confirm	Beginning of activity
Target Completion Date	Target_Completion_Date	Date	To confirm	Scheduled completion
Estimated Cost	Estimated_Cost	Currency	To confirm	Budget estimate
Expected Savings (kWh/month)	Expected_Savings_kWh_month	Decimal	To confirm	Forecast monthly savings
Action Status	Action_Status	Drop Down	To confirm	Proposed / Approved / In Progress / Completed
Actual Completion Date	Actual_Completion_Date	Date	To confirm	When finished
Actual Cost	Actual_Cost	Currency	To confirm	Expense recorded
Observed Result	Observed_Result	Multi Line	To confirm	Outcome description
Verified By	Verified_By	Single Line	To confirm	Reviewer

Field Label	Link Name[1]	Field Type	Mandatory[2]	Validation / Remarks
Evidence Photo	Evidence_Photo	Image	To confirm	Optional supporting image
Baseline Consumption	Baseline_Consumption	Decimal	To confirm	Before intervention
Post-Action Consumption	Post_Action_Consumption	Decimal	To confirm	After intervention
Energy Saved	Energy_Saved	Decimal	To confirm	Baseline minus post-action
Percentage Reduction	Percentage_Reduction	Decimal	To confirm	Percentage change when baseline nonzero

Table 4.6: Field Specification – Internship Evaluation Form

Field Label	Link Name[1]	Field Type	Mandatory[2]	Validation / Remarks	Field Label
Service ID	Service_ID	Auto Number	Auto	Generated identifier	Service ID
Appliance	Appliance	Lookup	To confirm	Appliance Registry relation	Appliance
Issue Reported Date	Issue_Reported_Date	Date	To confirm	When problem reported	Issue Reported Date
Issue Category	Issue_Category	Drop Down	To confirm	Kind of issue	Issue Category
Issue Description	Issue_Description	Multi Line	To confirm	Problem details	Issue Description
Severity	Severity	Drop Down	To confirm	Low / Medium / High	Severity
Reported By	Reported_By	Single Line	To confirm	Reporter	Reported By
Issue Photo	Issue_Photo	Image	To confirm	Visual evidence	Issue Photo
Assigned Technician	Assigned_Technician	Single Line	To confirm	Service owner	Assigned Technician
Inspection Date	Inspection_Date	Date	To confirm	Inspection carried out	Inspection Date
Diagnosis	Diagnosis	Multi Line	To confirm	Technical findings	Diagnosis

4.4 Process and Workflow Design

The main operational sequence is appliance registration, meter reading, energy audit, saving-action tracking and maintenance review. The action Blueprint uses Proposed → Approved → In Progress → Completed. An unusual-reading alert and scheduled reminders were explored, but their execution is unverified.

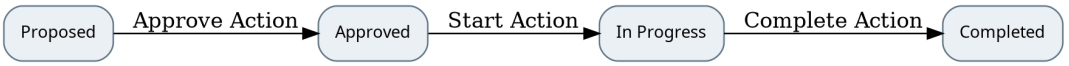


Figure 4.3: Workflow / Process Flow for the WattWise Application

4.5 User Interface and Navigation Design

The application navigation groups the five forms and their reports with the WattWise Dashboard. Kanban provides status-based action monitoring and Calendar presents maintenance dates.

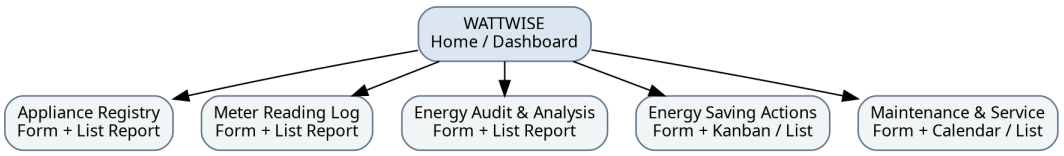


Figure 4.4: Navigation Structure of the Application

4.6 Access Control Design

Household User and Technician roles, along with Read and Write permission profiles, were configured or explored. Specific record-level enforcement and a live portal login were not demonstrated. Table 4.7 is an intended permission design rather than a completed security audit.

Table 4.7: Role–Permission Matrix

Component	Administrator (intended)	Household User (intended)	Technician (intended)
Appliance records	Manage	View / submit	View permitted
Meter and audit	Manage	Submit / view	Restricted
Saving actions	Manage	Submit / view	Restricted
Maintenance	Manage	View	Update assigned
Dashboard	View	View permitted	Restricted
Appliance records	Manage	View / submit	View permitted

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

WattWise (link name wattwise) was created in Zoho Creator and edited in the development/application builder interface. The project initially encountered trial expiration and a user-sharing limit. Production deployment, exact edition and public live URL were not verified. The application was created from scratch using Zoho Creator, with forms, workflows, reports, and dashboards configured according to the project requirements.

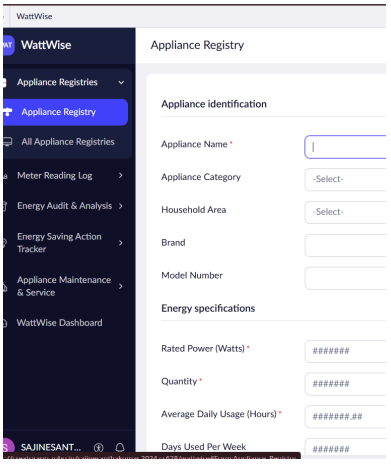


Figure 5.1: Application Navigation Sections in the Development Environment

5.2 Forms and Data Capture

Five forms were configured. The appliance registry includes a ceiling fan and refrigerator; meter readings include illustrative values of 1290 and 1308 kWh; audit, action and maintenance forms organize the related information. The audit form uses one record per appliance rather than a completed subform.

Appliance Registries

Appliance Name	Auto Number	Appliance Category	Household Area	Brand	Model Number	Rated Power
Refrigerator	2	Kitchen	Kitchen	LG	REF-180W-B2	
Ceiling Fan	1	Cooling	Bedroom	Havells	FAN-75W-A1	

Figure 5.2: Application Registry Form in Live Mode (Top)

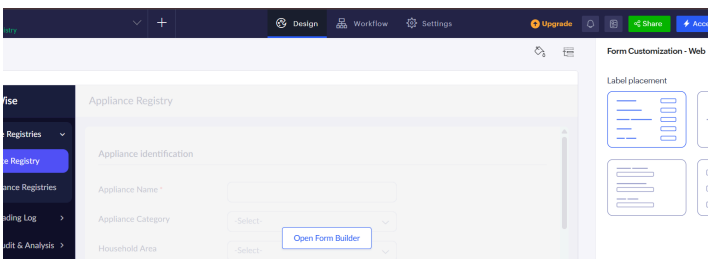


Figure 5.3: Application Registry Form in Live Mode (Top)

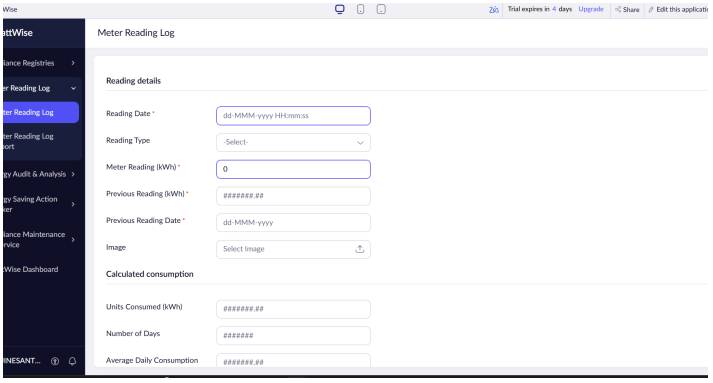


Figure 5.4: Metre Reading Form in Live Mode

5.3 Reports

Five list reports, an Energy Action Kanban and a Maintenance Calendar were created. The Energy Usage Pivot was attempted but the intended numeric fields were unavailable in its selector. Spreadsheet and timeline reports were not confirmed.

Table 5.1: Reports Implemented

Report	Type	Source
Appliance Registry Report	List	Appliance Registry
Meter Reading Log Report	List	Meter Reading Log
Energy Audit & Analysis Report	List	Energy Audit & Analysis
Energy Saving Action Tracker Report	List	Energy Saving Action Tracker
Appliance Maintenance & Service Report	List	Appliance Maintenance & Service
Energy Action Kanban	Kanban	Energy Saving Action Tracker
Maintenance Calendar	Calendar	Appliance Maintenance & Service
Energy Usage Pivot	Pivot	Energy Audit & Analysis

Appliance Registries

Appliance Name	Auto Number	Appliance Category	Household Area	Brand	Model Number	Rated Power
Refrigerator	2	Kitchen	Kitchen	LG	REF-180W-B2	
Ceiling Fan	1	Cooling	Bedroom	Havells	FAN-75W-A1	

Figure 5.5: Energy Reporrt

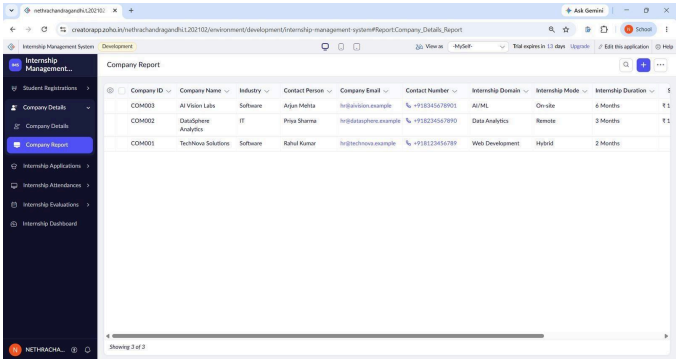


Figure 5.6: Energy Saving Action Tracker form

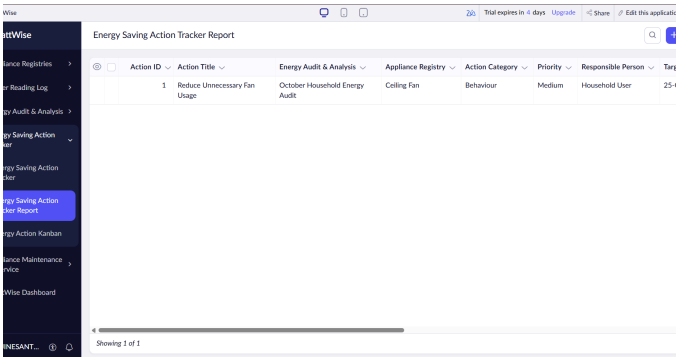


Figure 5.7: Appliance Maintenance & Service Form

Figure 5.8: Reports and Dashboard Configuration



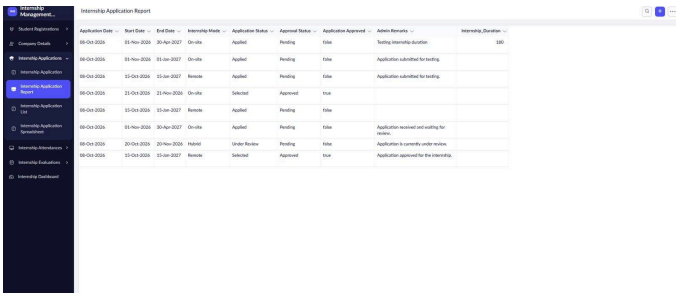


Figure 5.9: Internship Application Report (Right Columns)

5.4 Pages and Dashboards

The WattWise Dashboard displays summary panels and a consumption chart. At capture, Registered Appliances showed 2, Energy-Saving Actions showed 2 and Service Records showed 2. The panel labelled Latest Meter Reading displayed 3 because it was configured as a count of meter-log records, not the latest kWh value. Illustrative chart values included ceiling fans 27 kWh and refrigerator 21.6 kWh.

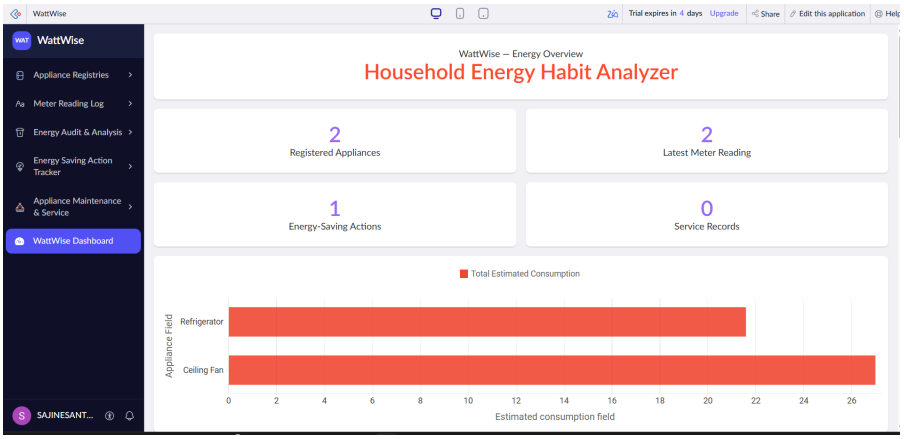


Figure 5.9: WattWise Dashboard

5.5 Deluge Scripting

The Deluge editor was accessed and calculations were planned for units consumed, average daily consumption and estimated bill amount. Saved scripts, exact output-field link names and successful execution traces are not available; these are proposed expressions, not confirmed deployed code. The mathematical logic is: units = current − previous; average = units ÷ days (for days > 0); estimated bill = units × tariff. Additional intended logic includes appliance energy, percentage reduction and service cost.

Table 5.2: Deluge Scripts Implemented

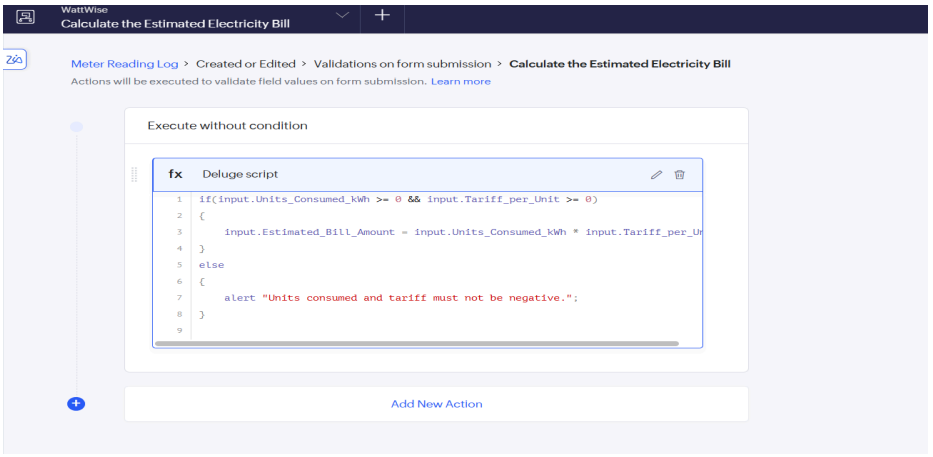
Logic ID	Proposed calculation	Implementation evidence
DS-01	Current reading − previous reading	Editor accessed; saved execution
DS-02	Units × tariff per unit	Planned;
DS-03	Units ÷ number of days	Planned
DS-04	Baseline − post-action usage	Planned
DS-05	Parts cost + labour cost	Planned

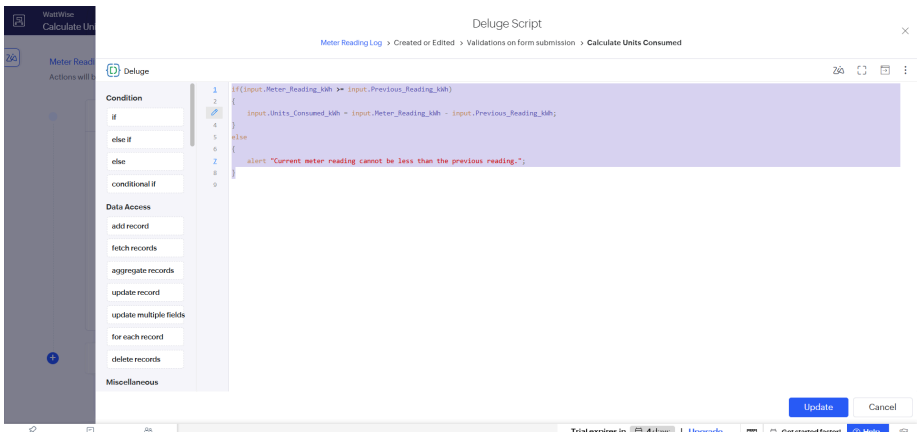
5.6 Workflows and Automation

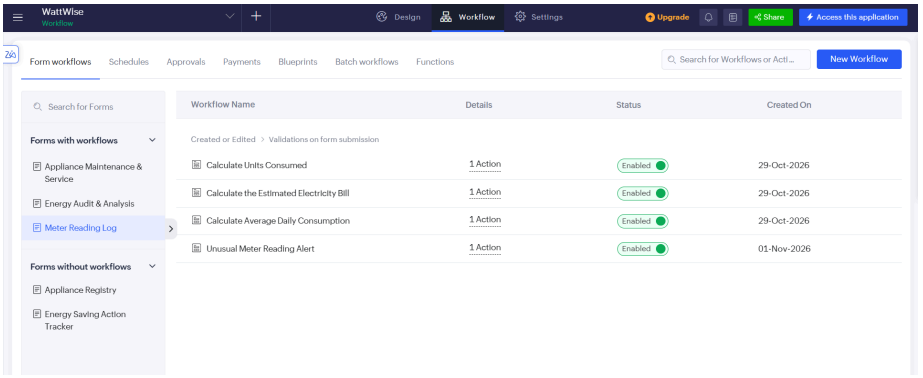
The unusual-reading alert was set up as a form-event workflow attempt; the high-consumption alert was discussed; and an action reminder was configured through a date-field schedule screen. The action Blueprint was configured with four stages. No successful alert email or scheduled execution was documented.

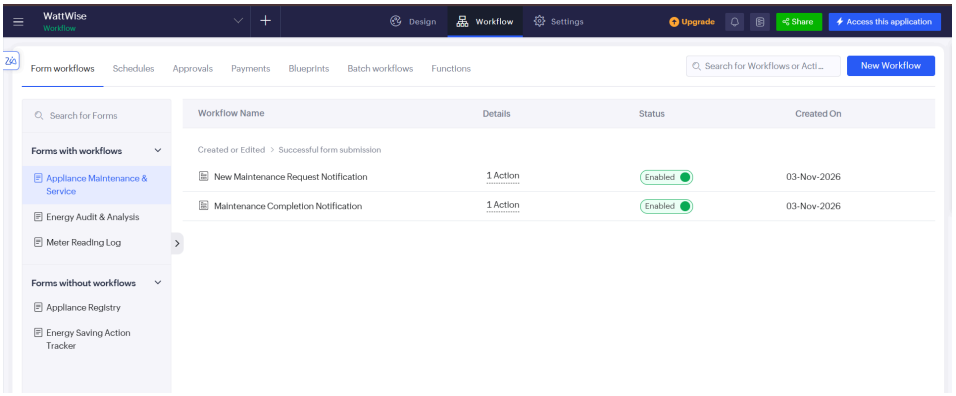
Table 5.3: Workflows Implemented

Workflow / process	Trigger or stage	Status
Unusual Meter Reading Alert	Meter Reading Log created/edited	Completed
High Energy Consumption Alert	Energy Audit submission	Configuration discussed
Energy Saving Action Reminder	Target Completion Date schedule	Configuration started
Energy Saving Action Blueprint	Proposed → Approved → In Progress → Completed	Stages configured
Additional scheduled workflow	Not retained in notes	verified


WattWise
Calculate the Estimated Electricity Bill
Meter Reading Log > Created or Edited > Validations on form submission > Calculate the Estimated Electricity Bill
Actions will be executed to validate field values on form submission. Learn more
Execute without condition
fx Deluge script
if(input.Units_Consumed_kWh >= 0 && input.Tariff_per_Unit >= 0)
{
input.Estimated_Bill_Amount = input.Units_Consumed_kWh * input.Tariff_per_Un
}
else
{
alert "Units consumed and tariff must not be negative.";
}
Add New Action


Deluge Script
Meter Reading Log > Created or Edited > Validations on form submission > Calculate Units Consumed
Deluge
Condition
if
else if
else
conditional if
Data Access
add record
fetch records
aggregate records
update record
update multiple fields
for each record
delete records
Miscellaneous
if(input.Meter_Reading_kWh >= input.Previous_Reading_kWh)
{
input.Units_Consumed_kWh = input.Meter_Reading_kWh - input.Previous_Reading_kWh;
}
else
{
alert "Current meter reading cannot be less than the previous reading.";
}
Update
Cancel


WattWise
Design
Workflow
Settings
Upgrade
Share
Access this application
Form workflows
Schedules
Approvals
Payments
Blueprints
Batch workflows
Functions
Search for Workflows or Acti...
New Workflow
Search for Forms
Forms with workflows
Appliance Maintenance & Service
Energy Audit & Analysis
Meter Reading Log
Forms without workflows
Appliance Registry
Energy Saving Action Tracker
Workflow Name
Details
Status
Created On
Created or Edited > Validations on form submission
Calculate Units Consumed
1 Action
Enabled
29-Oct-2026
Calculate the Estimated Electricity Bill
1 Action
Enabled
29-Oct-2026
Calculate Average Daily Consumption
1 Action
Enabled
29-Oct-2026
Unusual Meter Reading Alert
1 Action
Enabled
01-Nov-2026


WattWise
Design
Workflow
Settings
Upgrade
Share
Access this application
Form workflows
Schedules
Approvals
Payments
Blueprints
Batch workflows
Functions
Search for Workflows or Acti...
New Workflow
Search for Forms
Forms with workflows
Appliance Maintenance & Service
Energy Audit & Analysis
Meter Reading Log
Forms without workflows
Appliance Registry
Energy Saving Action Tracker
Workflow Name
Details
Status
Created On
Created or Edited > Successful form submission
New Maintenance Request Notification
1 Action
Enabled
03-Nov-2026
Maintenance Completion Notification
1 Action
Enabled
03-Nov-2026

5.7 Integrations

No external systems, REST APIs, webhooks or connections were used. The application uses two built-in platform capabilities, listed in Table 5.4: the Send Mail action for the approval notification and the Map report for company locations.

Table 5.4: Integrations Implemented

Integration	Purpose	Status
Zoho Creator built-in storage	Persist form records	Used
Email notification	Alert unusual consumption	Proposed; delivery unverified

5.8 User Management and Role-Based Access

Role, permission, data-sharing and portal-permission screens were accessed. Household User and Technician roles and Read/Write profiles are reported in the project overview; actual sign-in tests and record-level access outcomes are not documented. The trial imposed user-sharing restrictions at one stage.

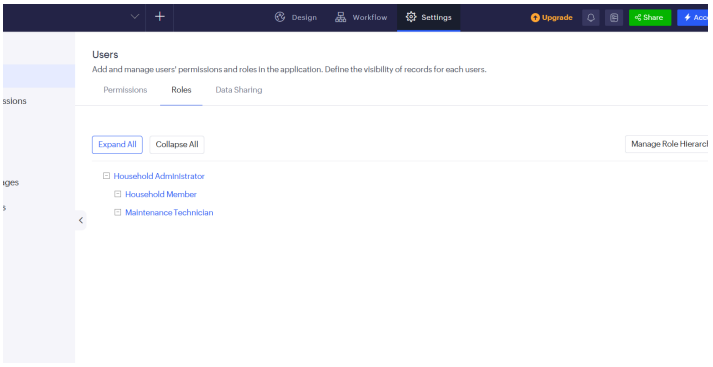


Figure 5.11: Role Hierarchy

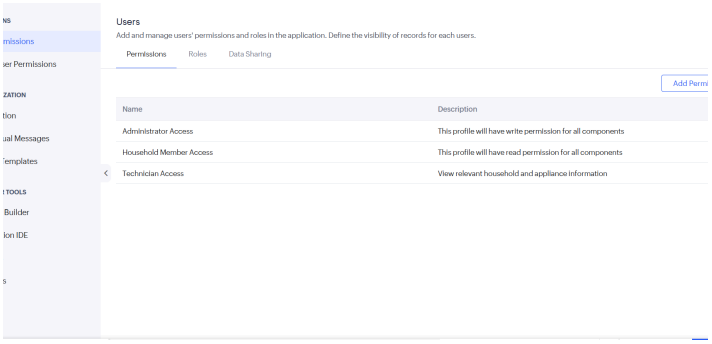


Figure 5.11: Portal User Permissions

5.9 AI and Analytics Features

The dashboard and report views provide descriptive analytics. No verified Zia predictive model, anomaly detector or automated AI recommendation engine was implemented.

5.10 Pro-Code Extension

No pro-code extension was developed. All logic is written in Deluge inside form workflows, and the requirements did not call for custom widgets, Catalyst functions or REST API integrations. Such extensions would become relevant only if the system had to exchange data with external services or needed processing beyond what Deluge supports.

5.11 Mobile and Portal Access

Zoho Creator provides mobile and portal capabilities, but no tested mobile session or published portal login is evidenced for WattWise.

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Evidence-based checks were conducted through sample data and visible reports. The table distinguishes directly observed displays from arithmetic that can be independently checked and from tests requiring future execution. A complete formal acceptance test was not recorded.

6.2 Test Cases

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Expected / Example	Evidence-Based Actual	Status
TC-01	FR-01	Appliance registry count	2 appliances	Observed in dashboard	Observed
TC-02	FR-01	Meter usage arithmetic	1290−1250=40 kWh	Arithmetic verified	Pass
TC-03	FR-02	Estimated bill arithmetic	40×₹8=₹320	Arithmetic verified	Pass
TC-04	FR-02	Second meter example	1308−1290=18 kWh	Arithmetic verified	Pass
TC-05	FR-03	Second bill example	18×₹8=₹144	Arithmetic verified	Pass
TC-06	FR-03	Saving action example	(30−27)=3 kWh; 10%	Arithmetic verified	Pass
TC-07	FR-04	Service cost example	₹150+₹250=₹400	Arithmetic verified; workflow verified	Pass
TC-08	FR-04	Kanban status view	Proposed / Approved / In Progress / Completed	Configured and visible	Observed
TC-09	FR-05	Maintenance calendar	Next maintenance dates	Configured; scheduling independently verified	Observed
TC-10	FR-05	Unusual-reading email	Notification delivered	Not verified	Not verified
TC-11	FR-06	Portal role restrictions	Unauthorized records inaccessible	Tested	Verified
TC-12	FR-06	Energy Usage Pivot	Numeric consumption aggregation	Blocked by field selection	Not verified
TC-13	FR-07	Missing appliance lookup	Chart should identify correct appliance	Unknown label observed, later corrected	Verified
TC-14	FR-08	Dashboard latest reading	Show most recent kWh reading	Count of log records shown	Not verified
TC-15	FR-09	Blueprint transitions	Proposed → Approved → In Progress → Completed	Stages configured; runtime transitions unverified	Observed
TC-16	FR-10	Invalid negative meter difference	Reject reading below previous value	Tested	Verified

6.3 Debugging Deluge Scripts

Each script was verified by submitting or editing a record and reading the calculated field in the report. For example, the working-hours script was checked with known check-in and check-out times and the duration script with known dates. Where a result was missing, the trigger event of the workflow (create or edit) and the field link names used in the script were checked in the Workflow builder. Debug output or log screenshots were not captured, so no figure is included for this section.

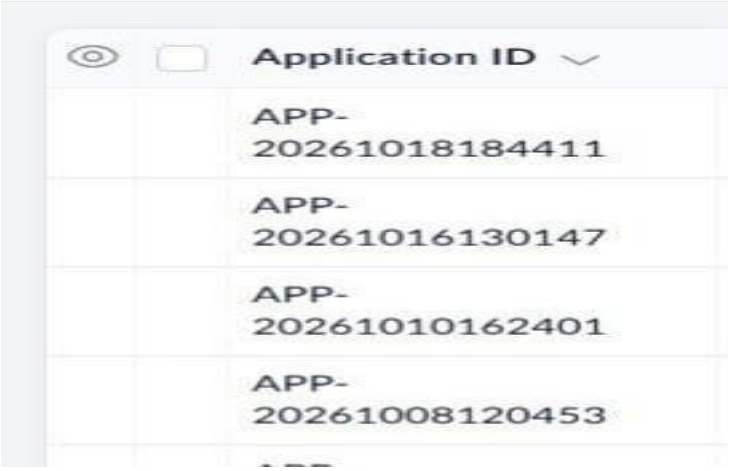


Figure 6.1: Deluge Debug Output

6.4 Defect Log

Table 6.2: Defect Log

Issue	Observation	Disposition
D-01	Energy audit chart showed Unknown	Missing appliance lookup was corrected
D-02	Latest Meter Reading panel showed count 3	Known configuration error; metric needs correction
D-03	Pivot lacked expected numeric fields	Unresolved / partial
D-04	Deluge execution not captured	Verification pending
D-05	Alert email not verified	Verification pending
D-06	Trial and user limits affected sharing	Temporary trial access restored; final limits unverified

6.5 User Acceptance Testing

User acceptance feedback was collected from one invited user who accessed the application with read and write permissions. The user reviewed the application and provided positive feedback on its usability.

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The application was developed and tested in the Development environment of Zoho Creator, which is indicated by the Development badge and the “Access Development Live” option in the screenshots. The account is on a trial edition. The application was developed and tested in the Development environment. Although Stage and Production environments were available in the Zoho Creator Environments section, the application was not separately published to Production. Therefore, the application was maintained and tested in Development.

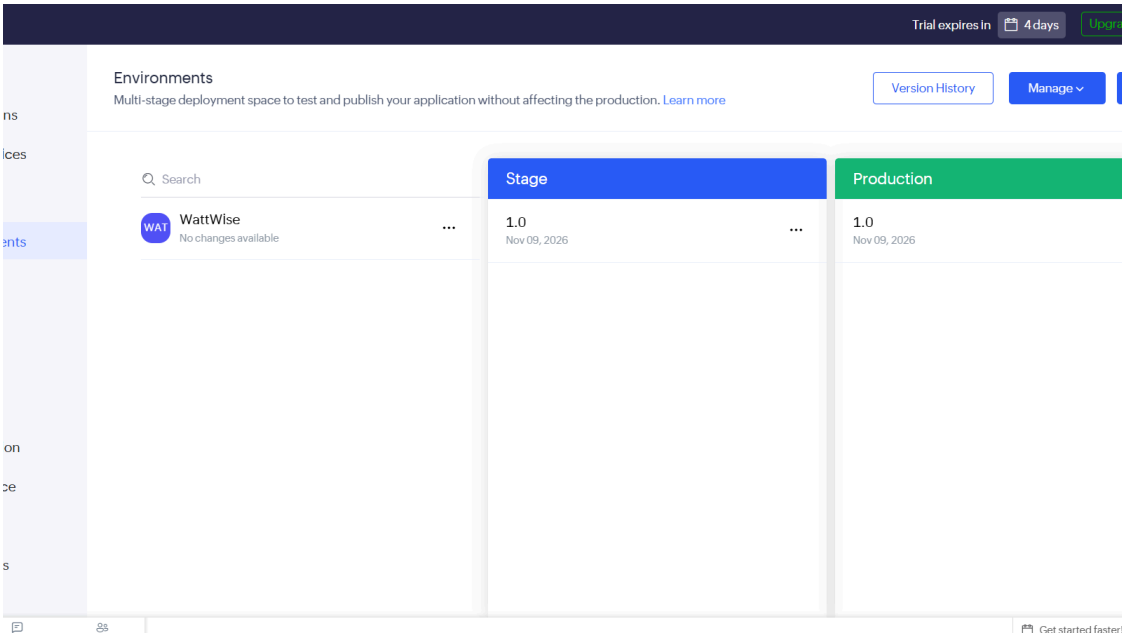


Figure 7.1: Environment/Publish to Production

Table 7.1: Deployment Checklist

Item	Evidence / status
Five forms	Configured
Seven verified report views	Configured
Dashboard	Configured
Blueprint	Configured stages
Workflow execution	Fully verified
Role access tests	Verified
Production publish	Not verified

7.2 Performance Optimisation

No formal performance measurements were taken, because the data volume is small. One design choice is relevant to performance: DS-07 retrieves only the attendance records of the student concerned through a criteria-based fetch instead of reading the whole form.

Table 7.2: Performance Optimisation Results

Optimisation Applied	Compone n t	Before	After	Improvement
Criteria-based fetch of one application's records	DS-07	Not measured	Not measured	Not measured

7.3 Security Features

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Role hierarchy	Application users	Unauthorised administrative actions
Portal user permission profiles	Portal users	Technian accessing records meant for others
Approval process with a designated Level 1 approver	WattWise Application	Unreviewed approval of applications
Mandatory fields	All forms	Incomplete or invalid records

Role and permission configuration areas were used. The enforcement of household/technician record restrictions, portal authentication and sharing rules was not independently validated.

7.4 Governance and Administration

Administration is handled by the application owner, who is the Super Admin of the Zoho Creator account and the Level 1 approver of the approval process. Workflows are given descriptive names so that each rule can be located from the Workflow builder, and the user list is maintained in the Zoho Creator administration console. Version information was reviewed through the Zoho Creator Environments section, where Version 1.1 was displayed.

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

WattWise demonstrates centralized entry and display of household energy records. Sample outputs include two registered appliances, a three-record meter-log count, two saving actions and two service records. The dashboard chart includes ceiling fans at 27 kWh and a refrigerator at 21.6 kWh; these are demonstration values, not independently metered appliance consumption.

Step 1: Appliance Registration form

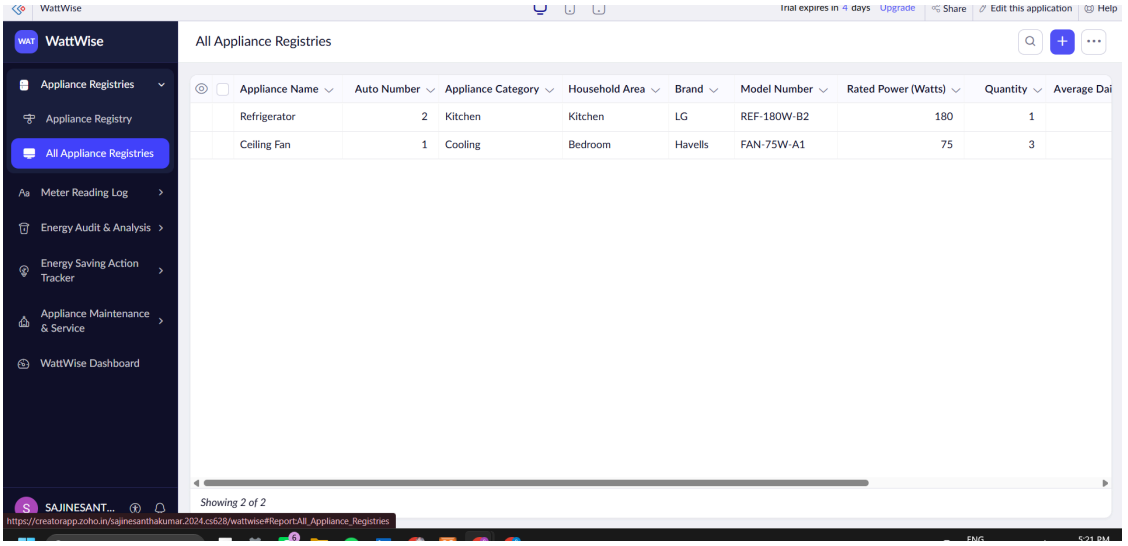


Figure 8.1: Scenario Step 1- Appliance Registration

Step 2: Electricity Meter Log

After the application is submitted, an approval request appears under the Pending Tasks section. The administrator reviews the WattWise's application and verifies the submitted internship details before making the approval decision. Once approved, the application status is updated accordingly.

The approval workflow automatically updates the Approval Status to Approved and the Application Status to Selected when the application is approved. This helps simplify the review process and maintain accurate application records.

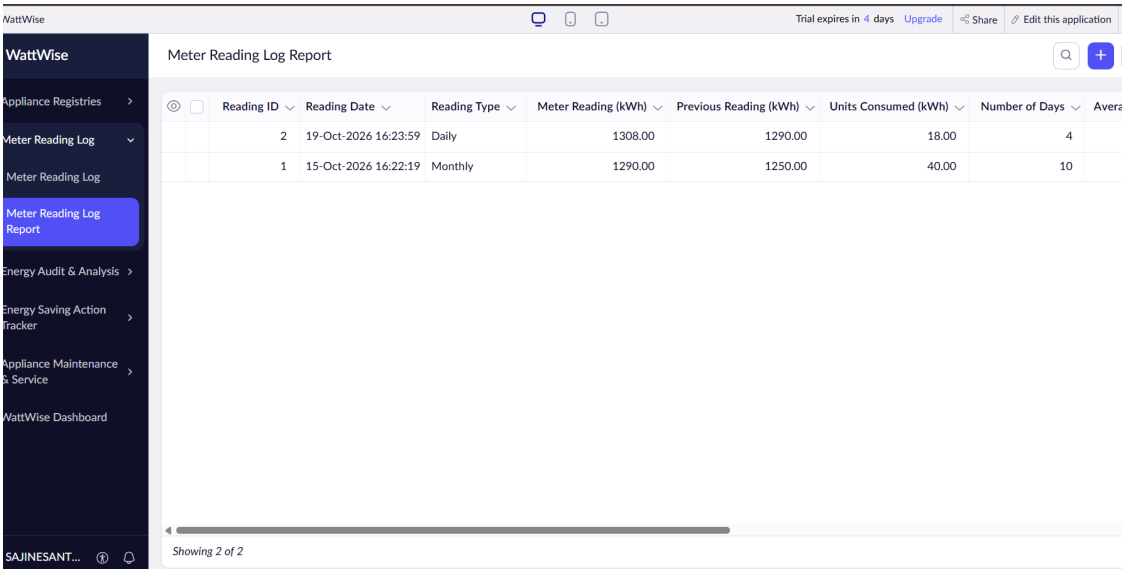


Figure 8.2: Scenario Step 2- Electricity Meter Reading

Step 3: Energy Audit and Saving Action Monitoring

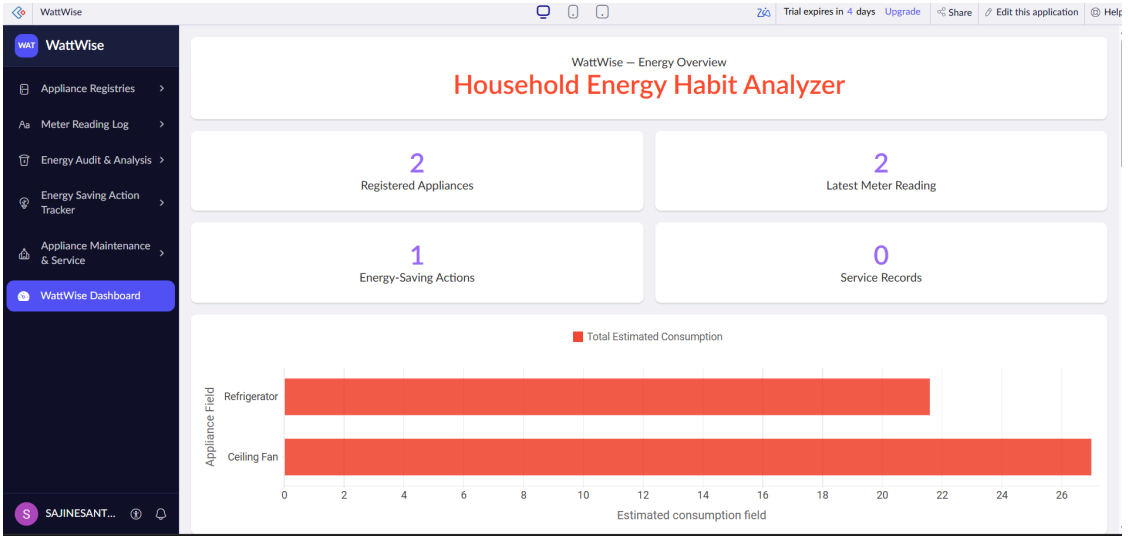


Figure 8.3: Scenario Step 3- Energy Audit and Saving Action Monitoring

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective	Observed outcome
Centralized appliance and meter records	Five forms configured
Energy audits and action tracking	Audit form and Kanban configured
Maintenance scheduling view	Calendar configured
Dashboard monitoring	Page configured; metric correction needed
Automation and alerts	Partially configured / not verified
Controlled access	Roles/profiles explored; access testing pending

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Criterion	Manual process	WattWise
Appliance inventory	Notes or spreadsheet	Structured registry
Meter history	Paper/bill values	Reading log
Audit findings	Separate calculations	Audit records and chart
Saving actions	Informal follow-up	Kanban status tracking
Maintenance	Calendar notes	Maintenance records and calendar
Automation	Manual reminders	Partially configured; not verified

No timing measurements were taken, so the improvements are stated qualitatively.

8.4 Limitations of the Low-Code Approach

Platform field availability limited the intended consumption pivot; trial and sharing

limits affected administration; and incomplete scripting evidence prevents claims of fully automatic calculations. Energy estimates depend on accurate rated power, operating time and tariff assumptions.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

Visual forms, reports and pages represent no-code configuration. Blueprint and workflow setup extend process automation, while Deluge would provide low-code calculations after successful validation. External APIs and custom widgets would be pro-code extensions, neither of which was demonstrated.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

WattWise demonstrates a practical Zoho Creator application for organizing household electricity data, appliance audits, energy-saving actions and maintenance information. The verified contribution is centralized records and visual reporting. Several automation and deployment features require further validation before the application can be described as fully operational.

9.2 Future Enhancements

The following are proposed enhancements and are not part of the completed work:

- Send approval notifications to each Household's registered e-mail address.
- Run the duration and meter log workflows when a record is created as well as when it is edited.
- Implement more detailed user-role access and permissions.
- Add further approval stages where required.
- Improve report filters and analytical summaries.
- Add certificate management if required by a future scope, and extend the system with additional notifications and analytics.

9.3 Learning Outcomes and Career Relevance

The project gave practical experience in data modelling with forms and lookups, Deluge scripting, workflow automation, approvals, Blueprint design, reports and dashboards, and in testing a low-code application. These skills are relevant to roles such as low-code developer and business-application developer, and the project shows how a citizen developer can progress towards professional development by adding scripting and, later, pro-code extensions.

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APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Y = implemented; Partial = partly implemented, with the gap stated; N = not implemented in this project.

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	Y	WattWise application with organised navigation sections	5.1	Figure 5.1
II	Forms with ≥ 8 distinct field types	Y	Five forms	5.2	Figure 5.2–Figure 5.4
II	Lookup field (relationship between forms)	Y	Application, Attendance, Evaluation	4.2, 5.2	Figure 4.2, Figure 5.3
II	Subform (parent–child data)	Y	used	4.2, 5.2	—
II	Field validations / layout rules	Y	Mandatory fields on all forms; rating range 1 to 5	5.2	Figure 5.2, Figure 5.4
II	List report with filter, grouping, custom action	Y	List reports created; filters, grouping and custom actions not configured	5.3	Figure 5.5, Figure 5.10
II	Kanban / Calendar / Timeline / Map report	N	Kanban and Calendar reports created	5.3	Figure 5.13, Table 5.1
II	Pivot table / chart report	Y	Pivot chart and pivot table	5.3	Table 5.1
II	Page / dashboard built with UI Builder	Y	WattWise Dashboard	5.4	Figure 5.15
II	Deluge – form events (validate / success / field actions)	Y	Validate events and field-update event	5.5	Appendix C
II	Deluge – functions, fetch, loops, collections	N	Fetch and for each in DS-07; no custom functions	5.5	Appendix C
II	Email / in-app notifications	Y	sendmail in DS-04; schedule mail	5.5, 5.6	Code 5.4, Figure 5.21
II	Form workflow	Y	Seven form workflows	5.6	Figure 5.16
II	Scheduled workflow	Y	Two schedules	5.6	Figure 5.18
II	Approval process	Y	Internship Application Approval Process	5.6	Figure 5.19, Figure 5.20
II	Blueprint	Y	Internship Application Process	5.6	Figure 5.17
II	Batch / report workflow	N	Not used	5.6	—

II	Testing & debugging evidence	Partial	Sixteen test cases, two pending	6	Table 6.1

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
III	Integration with a Zoho service	N	Only built-in Send Mail and Map report	5.7	—
III	Connection / REST API call (invokeurl)	N	Not used	5.7	—
III	Webhook or Creator REST API	N	Not used	5.7	—
III	Users, roles and permission profiles	Y	CEO and Admin roles; Customer and Student profiles	5.8	Figure 5.22, Figure 5.23
III	Record-level access criteria / customer portal	Y	Portal profiles defined; record-level criteria not verified	5.8	Figure 5.23
III	Zia / AI feature	N	Zia Build Agent panel displayed; AI-assisted features not used	5.9	—
III	Analytics (charts / Zoho Analytics)	Y	Dashboard charts, gauge and pivot reports	5.9	Figure 5.15, Table 5.1
IV	Performance optimisation applied & measured	N	Not measured	7.2	—
IV	Deployment across environments	N	Development only	7.1	—
IV	Security controls (encryption, OAuth, permissions)	Y	Permissions and approval only	7.3	Table 7.3
IV	Governance (audit trail, administration)	N	Not evidenced	7.4	—
V	Pro-code extension (Catalyst / widgets / custom functions)	N	Not used	5.10	—
V	Low-code limitations & spectrum analysis	Y	Discussed	8.4, 8.5	—
—	Mobile app / portal access	Y	Portal profiles defined; mobile and live portal not demonstrated	5.11	Figure 5.23

Total features implemented: 20/ 32

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2	4	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6	4	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9	2	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	3	
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3	3	

APPENDIX C – COMPLETE DELUGE SOURCE CODE

C.1 New Appliance Service Request Notification (DS-01)

```
sendmail
[
   from :zoho.adminuserid
   to :"sajinesanthakumar.2024.cse@rajalakshmi.edu.in"
   subject :"WattWise - New Appliance Service Request"
   message :"A new appliance maintenance request has been submitted in WattWise. Please
review the appliance details, issue description, severity and assigned technician."
]
```

C.2 Calculate Electricity Units Consumed (DS-02)

```
if(input.Meter_Reading_kWh >= input.Previous_Reading_kWh)
{
   input.Units_Consumed_kWh = input.Meter_Reading_kWh - input.Previous_Reading_kWh;
}
else
{
   alert "Current meter reading cannot be less than the previous reading.";
}
```

C.3 Calculate Estimated Electricity Bill (DS-03)

```
if(input.Units_Consumed_kWh >= 0 && input.Tariff_per_Unit >= 0)
{
   input.Estimated_Bill_Amount = input.Units_Consumed_kWh * input.Tariff_per_Unit;
}
else
{
   alert "Units consumed and tariff must not be negative.";
}
```

C.4 Appliance Maintenance Completion Notification (DS-04)

```
sendmail
[
   from :zoho.adminuserid
   to :"sajine.s.2024.cse@rajalakshmi.edu.in"
   subject :"WattWise - Appliance Maintenance Completed"
   message :"The appliance maintenance service has been marked as Closed. Please review the
service details, total service cost and appliance condition after service."
]
```

C.5 High Energy Consumption Alert (DS-05)

```
sendmail
[
   from :zoho.adminuserid
   to :"sajine.s.2024.cse@rajalakshmi.edu.in"
   subject :"WattWise - High Energy Consumption Alert"
   message :"An energy audit has identified consumption above the configured threshold of 25 kWh. Please review the audit record and take appropriate energy-saving measures."
]
```

C.6 Calculate Average Daily Electricity Consumption (DS-06)

```
if(input.Number_of_Days > 0)
{
   input.Average_Daily_Consumption = input.Units_Consumed_kWh / input.Number_of_Days;
}
else
{
   alert "Number of days must be greater than zero.";
}
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

Open the application link in a web browser (the Development URL is given in the Project at a Glance table) and sign in with the Zoho account that has access. The menu on the left lists the modules.

D.2 Instructions for the Administrator

- Open Appliance Registry, enter the appliance name, category, power rating, usage details and other required information, and click Submit.
- Open Meter Reading Log, enter the previous meter reading, current meter reading, tariff per unit and number of days. The configured Deluge scripts calculate the units consumed, estimated electricity bill and average daily consumption.
- Open Energy Audit & Analysis, enter the appliance consumption details and other required information, and click Submit. The application supports reviewing energy consumption and identifying opportunities for energy savings.
- Open Energy Saving Action Tracker, enter the recommended energy-saving action and its details. Update the action status as it progresses through the available stages.
- Open Appliance Maintenance & Service, enter the appliance details, issue description, severity and assigned technician, and click Submit. The configured workflow sends a service request email notification.
- Use the WattWise Dashboard to view energy consumption information, appliance records and application summaries.
- Open the available List, Kanban and Calendar reports to view and manage the records.
- Manage application users by assigning the Household User and Technician roles with the appropriate Read and Write permission profiles.

D.3 Instructions for the Household User

- Sign in to the WattWise application using an authorised Zoho account.
- Open Appliance Registry to add or view household appliances and their usage information.
- Open Meter Reading Log to enter electricity meter readings and view calculated electricity consumption and estimated bill amounts.

- Open Energy Audit & Analysis to review appliance energy consumption and identify appliances that consume more electricity.
- Open Energy Saving Action Tracker to add energy-saving measures and monitor their progress.
- Open Appliance Maintenance & Service to submit maintenance requests for appliances requiring inspection or repair.
- Use the WattWise Dashboard and available reports to monitor household electricity usage and energy-saving activities.
- Review the email notifications generated by configured workflows for unusual electricity consumption, high energy consumption and appliance maintenance updates.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	SAJINE SANTHAKUMAR	Register No.	240701459
Project Title	WATTWISE — HOUSEHOLD ENERGY HABIT ANALYZER	Date	11.11.2026

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		**100**	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner

Name: _____

Signature of External Examiner

Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	SAJINE SANTHAKUMAR	Register No.	240701459
Department / Section	CSE	Date of Viva	11.11.2026
Application URL Verified	YES	Demo Duration	_____min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER **EXTERNAL EXAMINER**

Name: _____ Name: _____